Repeat purchase prediction project checkpoint

Group 1 | Devano Jonathan | 3036029698 | 4 November 2026

Status. This checkpoint reports Group 1's progress on predicting whether an identified customer will make a valid purchase in the next 30 days from the previous 90 days of transactions. The current evidence supports continuing with the existing pipeline. Extended behavioural features improve PR-AUC over RFM, while the TF-IDF and MiniLM product-history features add little incremental value. My individual focus is the semantic product-feature track: reviewing cluster counts and labels, checking what MiniLM contributes, and explaining model errors.

Project progress

Dataset and design. The project uses the UCI Machine Learning Repository Online Retail dataset (ID 352), covering 541,909 invoice lines from 1 December 2010 to 9 December 2011. The analysis keeps only identified customers with a valid positive purchase. Month-end snapshots from March to October 2011 use a 90-day feature window and the following 30-day purchase outcome. March to June forms the training period, July selects the model, and August to October provides the final rolling test.

Data audit item	Value	Data audit item	Value
Raw invoice lines	541,909	**Missing customer IDs**	135,080
Exact duplicate rows	5,268	**Missing descriptions**	1,454
Cleaned valid purchase lines	392,692	**Positive identified invoices**	18,532
Customer-snapshot rows	16,155	**Modelled monthly cohorts**	8

Data decision. The one-year workbook remains the main source because every feature and label window is fully covered. Online Retail II does not add coverage for these dates: its 2010-2011 sheet is effectively the same data, and its two sheets repeat 22,523 rows from 1-9 December 2010.

Evidence from the current pipeline

Preliminary finding. The aligned pipeline document records PR-AUC results from the July validation and the pooled August-October evaluation. The main gain comes from richer behavioural features, not product text.

Model or baseline	July PR-AUC	Aug-Oct PR-AUC
Majority baseline	0.318	0.407
Recency rule	0.407	0.420
Logistic RFM	0.595	0.625
Boosting, behaviour	0.616	0.638
Boosting + TF-IDF	0.616	0.637
Boosting + MiniLM	0.615	0.638

Interpretation. Compared with RFM, extended behaviour raises pooled PR-AUC by about 0.009 for logistic regression and 0.008 for boosting. MiniLM changes PR-AUC by approximately 0.000 in both families. The July rule selected Boosting + TF-IDF, so the later pooled ranking is reported as a post-hoc comparison rather than used to replace the selection decision.

Individual focus and evidence

Devano Jonathan. My assigned track is Stages 8-9 and 30-32 in the group notebook. I am checking whether the product clusters are interpretable at K = 8, 12, 16 and 20, then using the July-selected model's October predictions to separate false positives from false negatives. This work explains why semantically meaningful product groups can still add little predictive value after behavioural features already capture recent activity and buying rhythm.

Concrete output. Planned evidence is a reviewed cluster-example table, a silhouette comparison for all four candidate cluster counts, an error-analysis table using final_test_predictions.csv, and a limitations section grounded in those results. The current workspace does not yet contain notebook_outputs, so the prediction file must be regenerated or shared before the error analysis can be completed.

Group contributions

Member	Work completed or in progress	Evidence or output	Approx. hours	Next checkpoint task
Stacia Benita	Auditing raw data and invoice construction; testing the effect of keeping versus dropping exact duplicates.	Data audit table, revenue reconciliation, duplicate sensitivity, two to three EDA plots, sourcing note.	Confirm	Finish audit evidence and record the cleaning decision by 6 November.
Devano Jonathan	Reviewing TF-IDF and MiniLM product clusters; checking K selection and interpreting October errors.	Named cluster examples, K = 8/12/16/20 silhouette table, error-analysis table, limitations draft.	Confirm	Finish cluster review and error analysis by 6 November.
Clara Natalie Surjadjajadi	Reproducing the v1 pipeline; checking leakage, baselines, July selection and rolling evaluation.	Verified notebook run, validation_results.csv, final_pooled_results.csv, leakage check and iteration log.	Confirm	Record baseline and leakage checks by 6 November.

Hours are not recorded in WORK_SPLIT.md. Each member should replace "Confirm" with their own approximate total before submission.

Next steps and blockers

Target date	Owner	Task
6 Nov 2026	All	Complete individual checkpoint evidence: data sensitivity, cluster and error review, and verified pipeline outputs.
7 Nov 2026	Each member	Submit an individual two-page checkpoint PDF with confirmed hours.
9-14 Nov 2026	Clara; Devano; Stacia	Integrate sensitivity and cluster-count results, finalise figures, and maintain the iteration log.
15-20 Nov 2026	All	Merge individual sections into the assessed report with one consistent argument and caveat set.
21-25 Nov 2026	All	Build slides, rehearse the story and run a final numerical consistency check.

Current blockers. The first blocker is the missing notebook_outputs folder, which prevents direct review of final_test_predictions.csv and cluster-example outputs. The second is that cluster names need human judgement rather than automatic acceptance. A cleaning change that affects snapshots would invalidate downstream results, so Stacia must report the duplicate sensitivity before Clara finalises the iteration log.

AI assistance and checks

AI was used to compare the two local workbooks, interpret the proposal and checkpoint requirements, and draft this document. The data counts were checked directly against the full workbooks, while the reported model results were cross-checked against the aligned pipeline document and notebook structure. Cluster labels, the final interpretation of MiniLM's small gain, member hours and submission wording still require human review by the group.